HotelRevAI
AI Driven Revenue Analysis for Hotels

Presented By:

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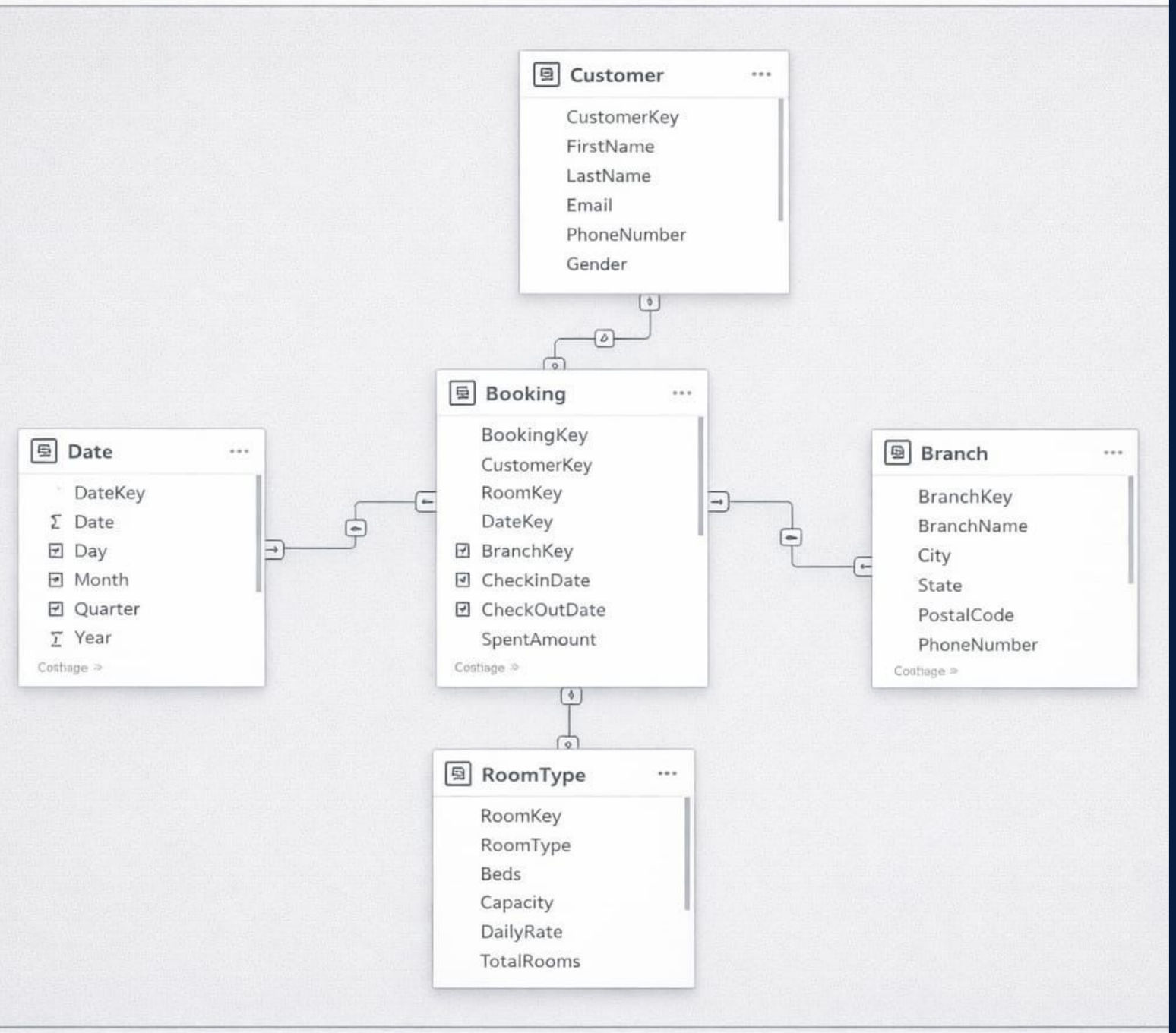
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RoomKey
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BranchKey
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Costiage
Date
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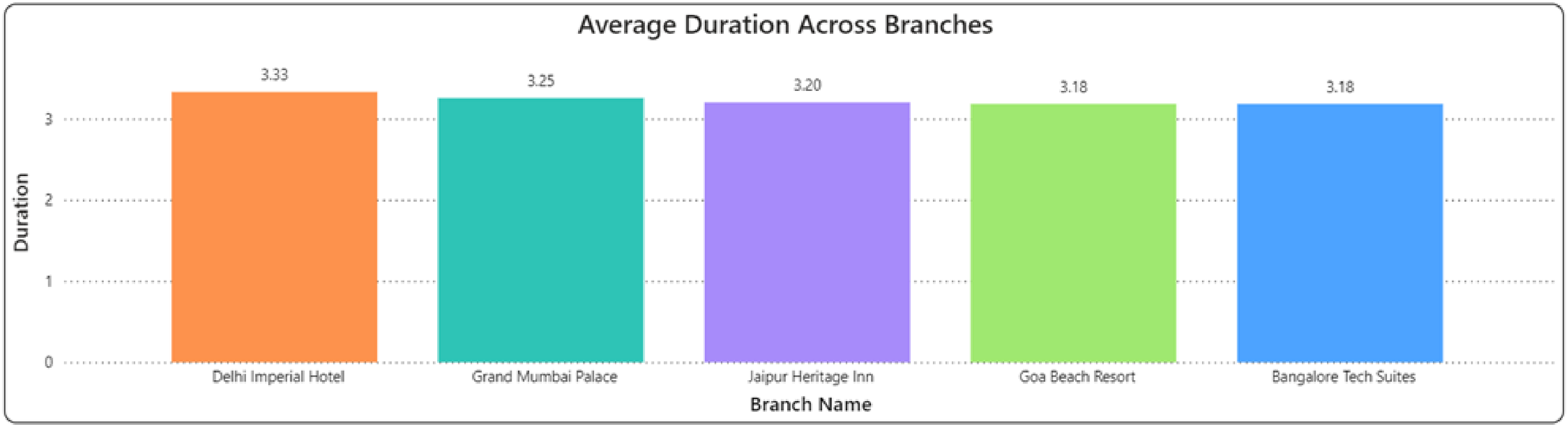
Date
Day
Month
Quarter
Year
Costiage
Branch
BranchKey
BranchName
City
State
PostalCode
PhoneNumber
Costiage
RoomType
RoomKey
RoomType
Beds
Capacity
DailyRate
TotalRooms

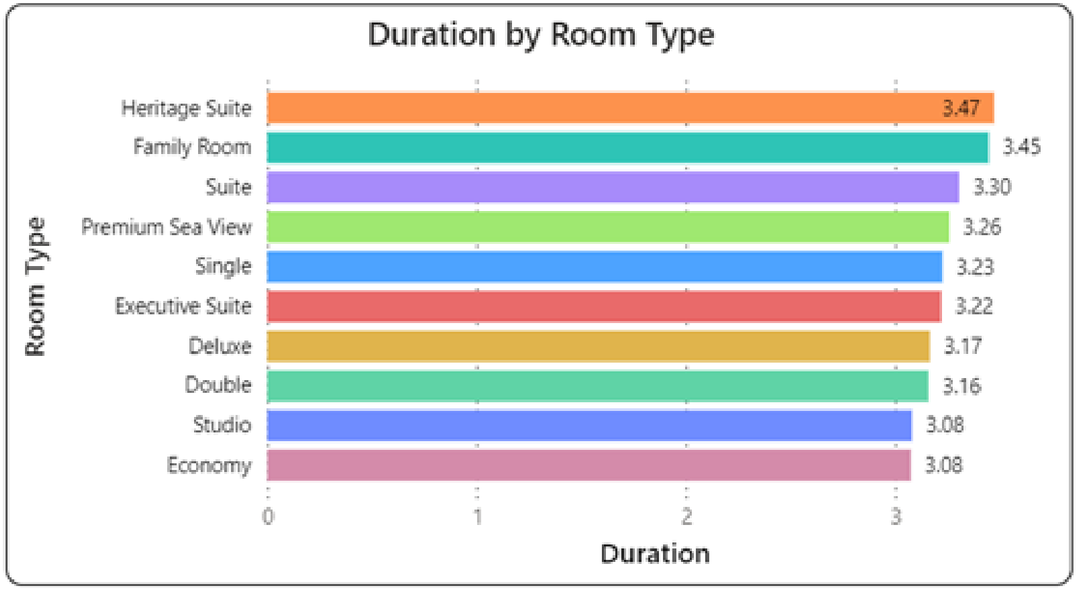
Star Schema

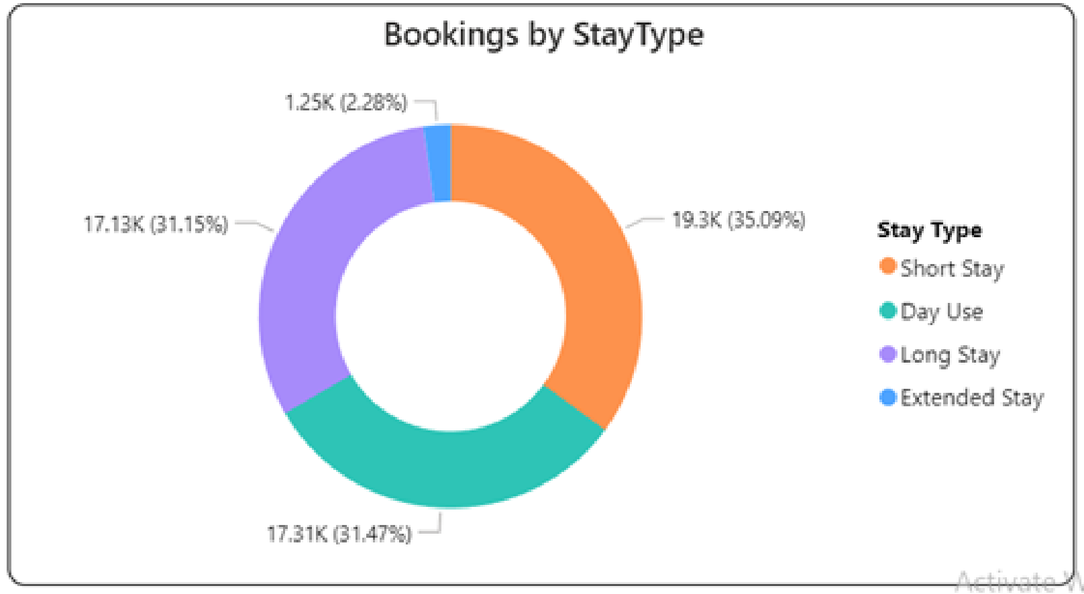
Module 1: DATA MODELING AND INGESTION

- Built scalable data foundation for revenue analytics
- Created realistic, high-volume hotel datasets with Python
- Standardized data into analytical-ready formats (CSV via Excel)
- Developed key derived attributes like Stay Type and Room Nights Sold
- Prepared data in Power BI for enterprise reporting and modeling

Booking Trends and Room Performance Analysis







MODULE 2: OCCUPANCY & REVENUE METRICS

KEY KPI

1 OCCUPANCY RATE

- % of rooms occupied
- Formula: (Rooms Booked / Total Rooms) × 100

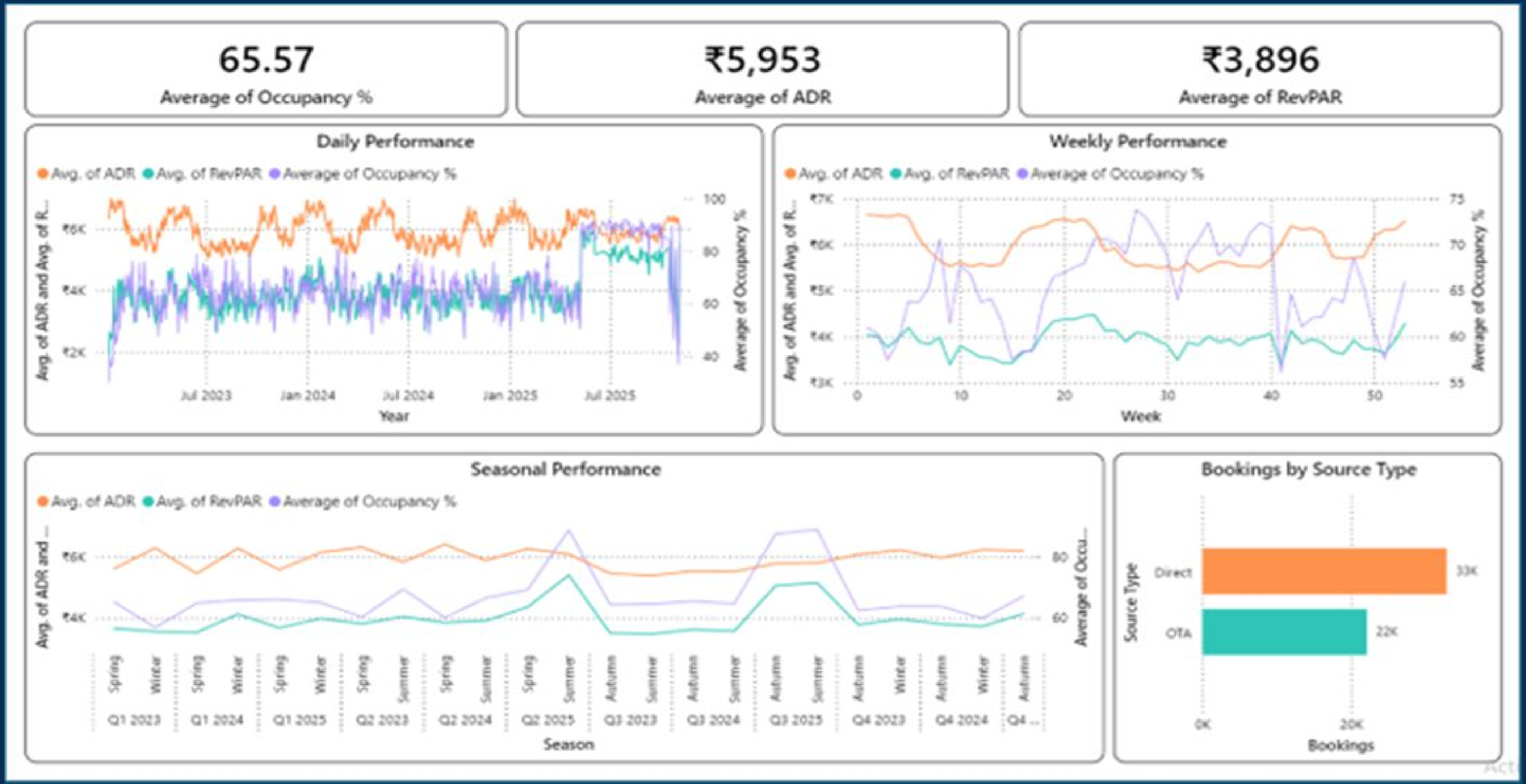
2 AVERAGE DAILY RATE (ADR)

- Revenue per booked room
- Formula: Total Revenue / Rooms Booked

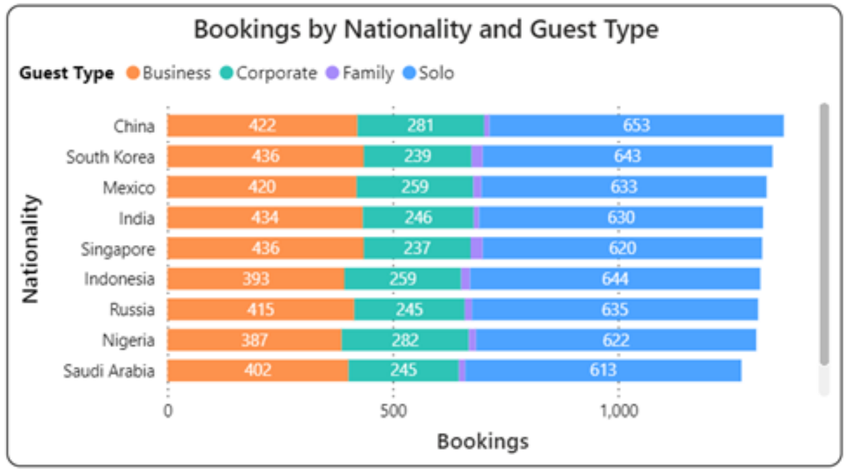
3 REVPAR (REVENUE PER AVAILABLE ROOM)

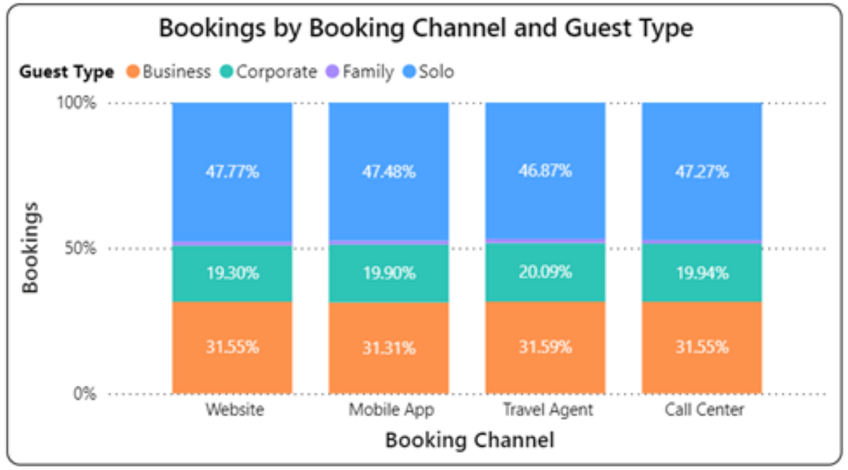
- Combines occupancy & pricing
- Formula: ADR × Occupancy Rate

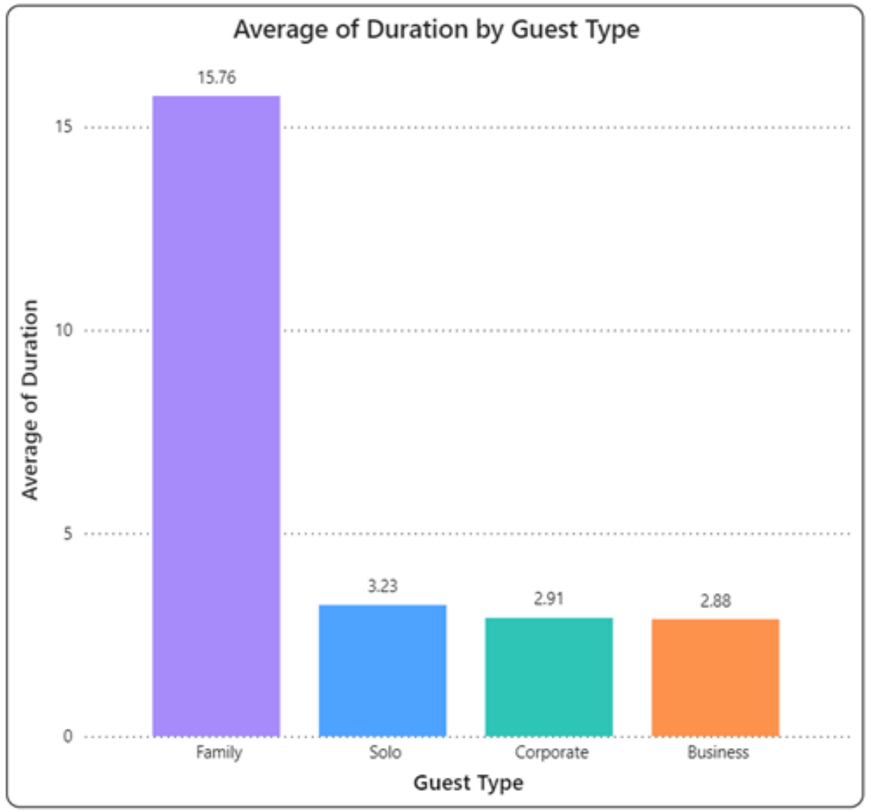
KEY INSIGHTS FROM DASHBOARD



MODULE 3 - GUEST ANALYSIS







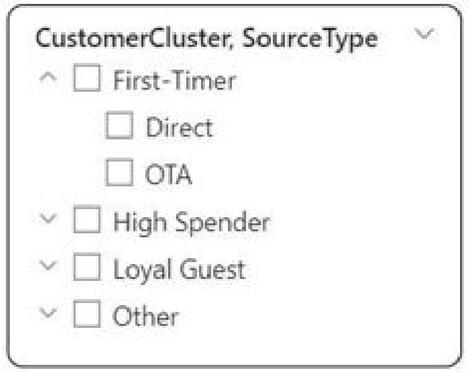
CustomerCluster Logic:

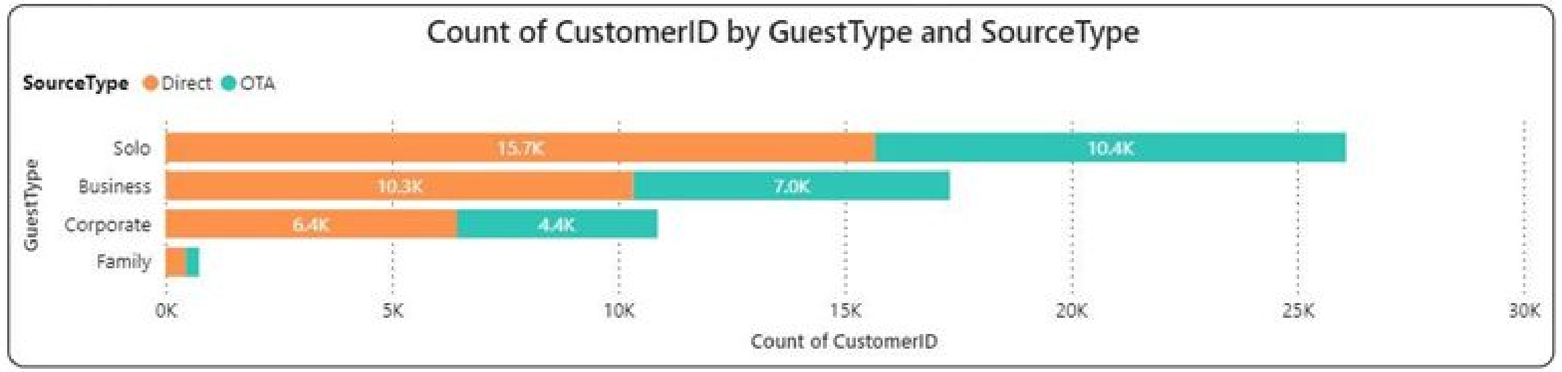
- 1 booking → First-Timer
- ≥ 3 bookings & low revenue → Loyal Guest
- High revenue → High Spender

GuestType Logic:

- Business → Corporate/Business purpose
- Family → Vacation/Extended stay
- Corporate → Conference/Long stay
- Solo → Default

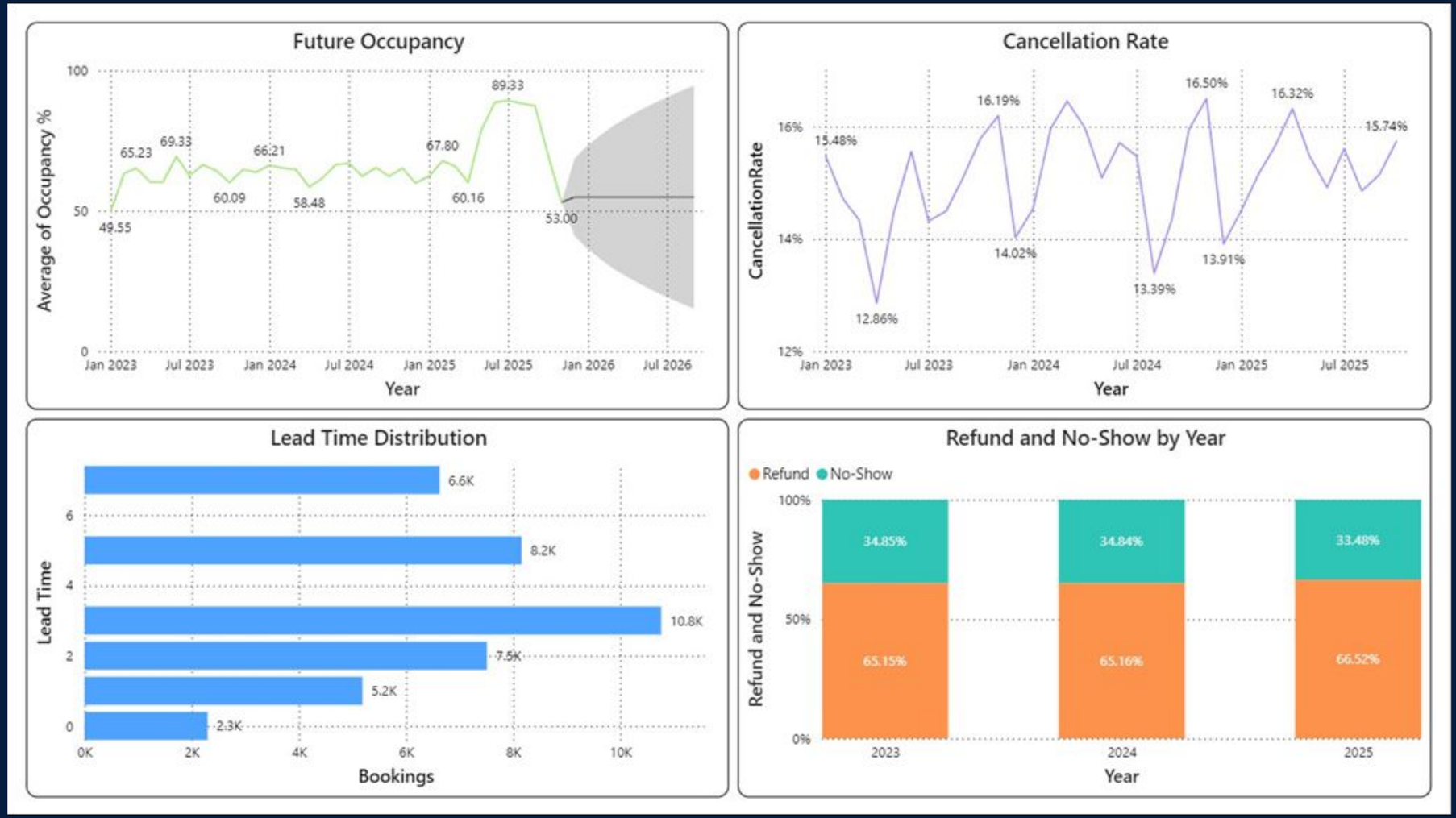
GUEST SEGMENTATION & DEMOGRAPHIC ANALYSIS





Module 4: Forecasting and Cancellation Trends

Key Insights & Business Value:



- Occupancy: Fluctuates between ~21% and ~36%, stabilizing near ~31% with uncertain future outlook.
- Cancellation Rate: Ranges from ~3% to ~37%, mostly staying between ~29%–36%.
- Lead Time: Majority of bookings fall within 0–60 days with consistent distributon.
- Refunds & No-Shows: Vary yearly (~0.4K–2.7K), with no-shows higher than refunds in recent years.

MODULE 5 – REVENUE STRATEGY DASHBOARD

Revenue Optimization Logic

Room Type Revenue Breakdown by Season

RoomType	High	Low	Shoulder	Total
Heritage Suite	11,500.00	9,000.00	10,500.00	**10,000.00**
Executive Suite	10,925.00	8,550.00	9,975.00	**9,500.00**
Suite	9,775.00	7,650.00	8,925.00	**8,500.00**
Premium Sea View	8,625.00	6,750.00	7,875.00	**7,500.00**
Family Room	8,050.00	6,300.00	7,350.00	**7,000.00**
Deluxe	6,900.00	5,400.00	6,300.00	**6,000.00**
Studio	6,325.00	4,950.00	5,775.00	**5,500.00**
Double	4,600.00	3,600.00	4,200.00	**4,000.00**
Single	2,875.00	2,250.00	2,625.00	**2,500.00**
Economy	2,070.00	1,620.00	1,890.00	**1,800.00**
Total	**7,125.28**	**5,204.31**	**6,024.35**	**5,897.35**

Occupancy Sensitivity

1.00

Price Uplift %

0.00

Customer Value Logic:

Higher ancillary spend → High-value customer

Frequent service usage → Upsell opportunity

Low ancillary usage → Target for promotions

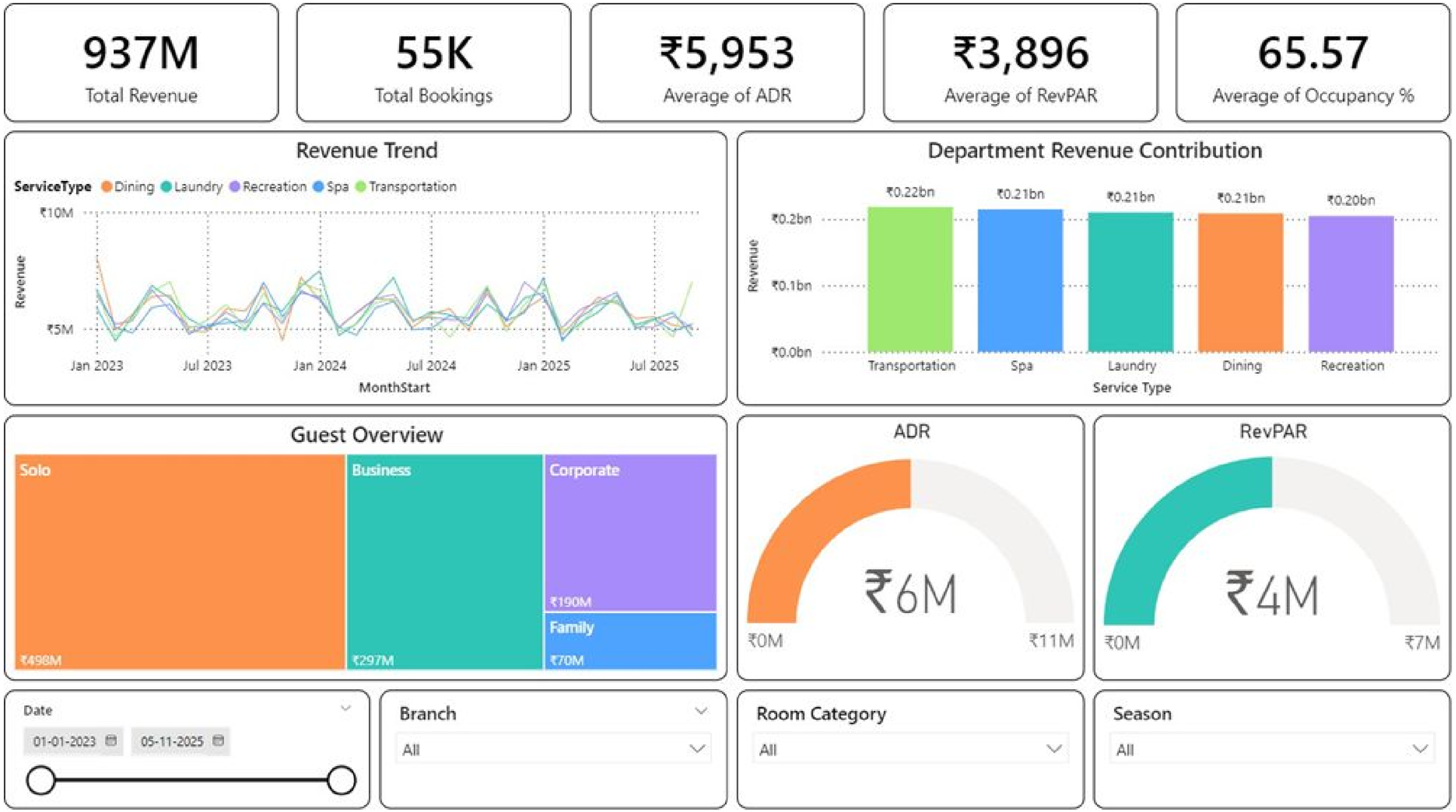
Pricing Strategy Logic:

High Season → Increase prices (High demand)

Low Season → Reduce prices (Boost occupancy)

Shoulder Season → Balanced pricing approach

General Manager Dashboard



Key Insights:

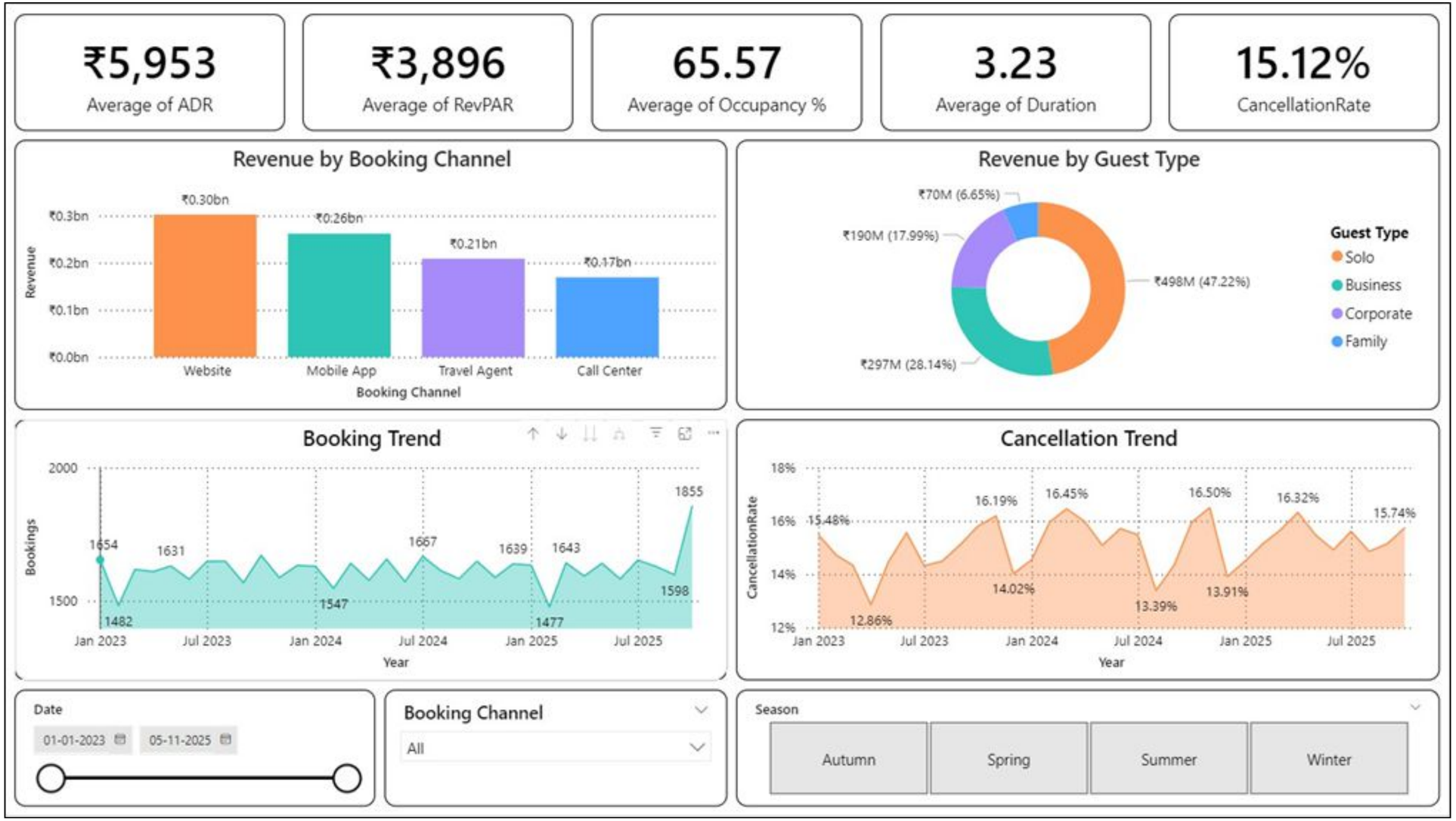
Overall Performance (Top KPIs)

Revenue Trends & Insights

Business Breakdown Analysis

Decision-Making Support

Revenue Manager Dashboard



Key Insights:

Lead Time Trend

KPI Overview

Time-Based Analysis

Performance Comparison

FUTURE ENHANCEMENTS

Machine Learning Models
- Advanced demand prediction

Dynamic Pricing
- Real-time price optimization

Customer Personalization
- AI-based recommendations

Real-Time Integration
- Live booking data

AI Chatbot
- Automated support

Conclusion

The HotelRevAI project helps analyze hotel performance using data and key metrics like Occupancy Rate, ADR, and RevPAR. It supports better decision-making, improves pricing strategies, and increases overall revenue. This solution makes hotel management smarter and more efficient.

Thank you